

# Erdős Problem #436

EP-436

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

## ORIGINAL PROBLEM STATEMENT

If  $p$  is a prime and  $k, m \geq 2$  then let  $r(k, m, p)$  be the minimal  $r$  such that  $r, r + 1, \dots, r + m - 1$  are all  $k$ th power residues modulo  $p$ . Let  $\Lambda(k, m) = \limsup_{p \rightarrow \infty} r(k, m, p)$ . Is it true that  $\Lambda(k, 2)$  is finite for all  $k$ ? Is  $\Lambda(k, 3)$  finite for all odd  $k$ ? How large are they?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>  
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>